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CHAPTER-1

INTRODUCTION

1.1 Background

Let's face it, taking attendance is a pain. Teachers have to stop what they're doing, call out names, wait for students to respond, maybe pass around a sheet. It's slow. It wastes time. And honestly, it just feels clunky. Sheets disappear, people can't read the handwriting, and students sign in for friends who aren't even in class. Proxy attendance is a headache for teachers and to keep up records, so teachers will not know who is coming to class may lead to proxy attendance.

Almost every classroom already has a computer or a laptop. So By adding a webcam and using today's computer vision tools, we tried we could fix the whole attendance system correctly. We don't need expensive equipment, we just need software.

1.2 Problem Overview

Manual attendance takes time, traditional roll number calling waste 5 to 10 minutes of class time. If you calculate that over a whole semester, it will cost hours of lost teaching time. Biometric systems like fingerprint scanners can be used, but they are expensive, slow, and not which may lead to diseases and virus spread since everyone touches the same thing.

We need some software and contactless a system that works quietly in the background, so the class gets moving. We needed a solution that does not force students to stand in queue not to waste time or touch a sensor, especially when hygiene is a priority.

1.3.0 Scope of the Work

We built a working prototype for a single classroom. Here's is what our system makes:

Snap student photos and build up a dataset. It takes about 30 seconds per student. register student details.

Training: “Teach” the computer to recognize these faces using the photos we just took.

1.3.1 Face Recognition Module

The system runs live during class and marks attendance automatically—no need for roll call.

1.3.2 Attendance Recording System

Once a student is recognized, the system automatically marks attendance and stores the details such as name, date, and time in a file or database.

1.3.3 Graphical User Interface

The system includes:

A simple GUI is provided to display the live camera feed, recognized student names, and attendance logs, making it easy for teachers to operate the system.

Outside the Scope

Large-scale institutional deployment

Mobile application development

Cloud-based storage and synchronization

Advanced spoof detection and surveillance features

CHAPTER-2

LITERATURE SURVEY

Sl. No.	Title and Year	Authors	Methodology	Technology Used	Gap / Drawbacks
1	Face Recognition Based Attendance System (2021)	R. Patel et al.	Used facial image comparison to automate attendance marking.	OpenCV, Python	Accuracy affected by lighting variations.
2	Automated Attendance Using Face Recognition (2022)	S. Kumar, A. Verma	Real-time face detection and recognition using webcam input.	OpenCV, Haar Cascade	Struggles with face occlusion.
3	Smart Attendance System Using AI (2023)	M. Singh et al.	Applied AI models for identifying students in classrooms.	Python, CNN	Requires high computational power.
4	Real-Time Attendance Monitoring System (2022)	J. Lee, K. Park	Compared live face images with stored datasets.	OpenCV, ML	Performance decreases with low camera quality.

5	AI-Based Attendance Management System (2024)	A. Sharma et al.	Used facial embeddings for attendance automation.	FaceNet, NumPy	Needs large datasets for better accuracy.
6	Automated Classroom Attendance System (2021)	P. Roy et al.	Detected faces and marked attendance automatically.	HOG, Python	Not suitable for crowded classrooms.
7	Contactless Attendance System Using Face Recognition (2023)	K. Nair, S. Mehta	Developed a contactless system to avoid manual attendance.	OpenCV, Deep Learning	Complex system setup.
8	Intelligent Attendance System for Education (2024)	T. Wang et al.	Implemented AI-based face matching for attendance.	TensorFlow, Python	High training time.
9	Student Attendance Using Facial Recognition (2022)	L. Fernandez et al.	Used facial feature extraction and matching techniques.	CNN, OpenCV	Sensitive to pose variations.
10	Smart Face Recognition Attendance System (2025)	S. Reddy et al.	Integrated face recognition with GUI-based attendance logs.	Python, Tkinter	GUI lag with high frame rates.
11	Automated Attendance System Using Computer Vision (2023)	D. Choudhary et al.	Computer vision-based face detection and recognition.	OpenCV, ML	Needs controlled environment.

12	AI-Powered Attendance Monitoring System (2024)	N. Gupta et al.	Used AI algorithms to recognize student faces in real time.	Deep Learning, Python	High hardware dependency.
13	Facial Recognition Attendance System for Colleges (2022)	H. Kim et al.	Developed a real-time recognition-based attendance tool.	OpenCV, FaceNet	Difficulties with identical twins.
14	Vision-Based Smart Attendance System (2023)	R. Malhotra et al.	Detected faces and marked attendance automatically.	CNN, OpenCV	Accuracy drops in poor lighting.
15	Attendance Automation Using AI and ML (2024)	P. Das et al.	Used ML classifiers for facial recognition.	Python, ML Libraries	Requires frequent retraining.
16	Face Recognition System for Student Attendance (2021)	S. Ali et al.	Compared facial features with pre-stored images.	Haar Cascade, OpenCV	Limited scalability.

17	AI-Based Attendance Tracking System (2025)	V. Iyer et al.	Combined real-time recognition with data storage.	Python, SQLite	Database performance issues.
18	Smart Attendance Using Deep Learning (2023)	A. Mehra et al.	Used deep learning models for face recognition.	CNN, TensorFlow	Computationally expensive.
19	Classroom Attendance Using Face Recognition (2022)	Y. Tanaka et al.	Implemented webcam-based real-time recognition.	OpenCV, Python	Not robust to motion blur.
20	Automated Attendance Management Using AI (2024)	R. Bose et al.	Integrated AI face recognition with attendance logging.	Python, OpenCV, GUI	Limited testing in large classrooms.

2.1 Critical Analysis of Literature :

Most of the studies seem to follow a similar pattern, where features are taken from the face first and then compared through embeddings.. CNN approaches usually give better accuracy, but they need stronger hardware, whereas HOG methods run faster and work well for real-time use. Researchers emphasize the need for lightweight models for edge devices. Studies show that Euclidean-distance-based similarity measures perform reliably for verification tasks. One issue that appeared often in the papers was maintaining real-time speed in GUI applications, and many researchers handled this by using parallel tasks or thread-based processing.

2.2 Implication & Conclusion :

From what the reviewed research shows, lighter face recognition models are capable enough to handle real-time attendance requirements. Using facial embeddings for recognition turns out to be both secure and easy to scale for classroom situations. The reviewed works support the feasibility of designing an attendance system based on facial encodings and threshold-based matching.

2.3 Problem statement:

Manual attendance takes time, traditional roll number calling waste 5 to 10 minutes of class time. If you calculate that over a whole semester, it will cost hours of lost teaching time. Biometric systems like fingerprint scanners can be used, but they are expensive, slow, and not which may lead to diseases and virus spread since everyone touches the same thing.

We need some software and contactless a system that works quietly in the background, so the class gets moving. We needed a solution that does not force students to stand in queue not to waste time or touch a sensor, especially when hygiene is a priority.

2.4 Objectives:

Our smart attendance system using AI have a objectives like :

- Build a system that runs on any regular laptop—no fancy servers, no pricey hardware. We focused on three main things:
- The camera picks up faces and marks attendance, no interruptions. The teacher doesn't even need to click a button once it starts.
- Since everyone's face is unique, students can not put proxy for each other.
- All the records go straight into a digital database, ready to export to Excel.

CHAPTER-3

REQUIREMENT ANALYSIS

Functional Requirements Specifications

1. User Authentication

The system will allow authorized users such as teachers and administrators to log in securely. This ensures that only users can access attendance data.

2. Student Registration

The system needs a registration module where you enter student details and snap their photos. It processes and stores these images for later.

3. Real-Time Face Detection

The system will detect student faces in real time using a webcam during classroom sessions.

4. Face Recognition and Matching

Detected faces will be checked with registered student data to identify students .

5. Automatic Attendance Logging

Once a student is recognized, attendance will be marked automatically along with date and time.

6. Attendance Report Generation

The system should allow users to generate attendance reports in format.

Software Requirements

- Python: Our main programming language. It's free and has great tools for images.
- OpenCV: For all the video and image processing. It handles the webcam feed.
- MediaPipe: Detects faces quickly without dragging down the computer. It works even on laptops.
- Tkinter: used to build desktop windows and buttons. It makes the app look like a real program.
- SQLite: Stores all the student info and attendance logs. It saves everything in a simple file.
- Flask: To run the web viewer so you can check lists on a mobile phone using Wi-Fi.

Hardware Requirements

- Laptop/PC: Anything with an Intel i3 processor or better.
- RAM: 4GB.
- Webcam: standard HD webcam (720p or 1080p).
- Hard Disk: We need about 500MB of space for the photos.

CHAPTER-4

SYSTEM DESIGN

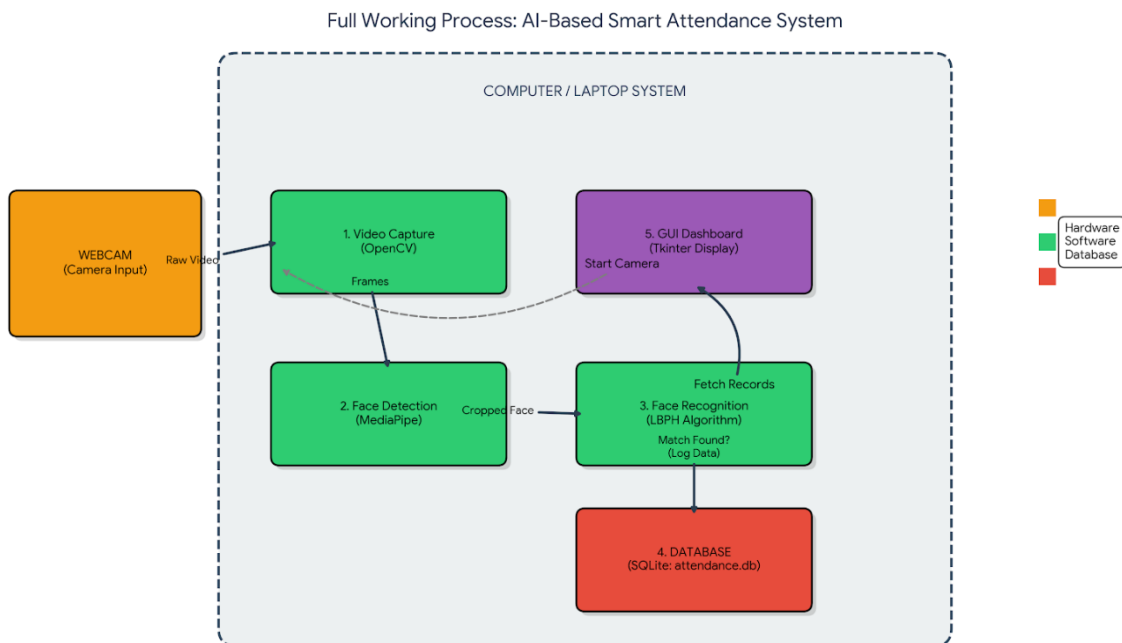
4.1 System Architecture :

We split the system into three simple parts:

Dashboard: This is what the teacher sees. We built it with Tkinter. It has big buttons for "Register," "Train," and "Start Attendance." It's designed like that one can use the system with easier .

Backend : This is the Python code doing the heavy lifting. We have scripts that grab photos, learn the patterns of the face, and run the recognition loop.

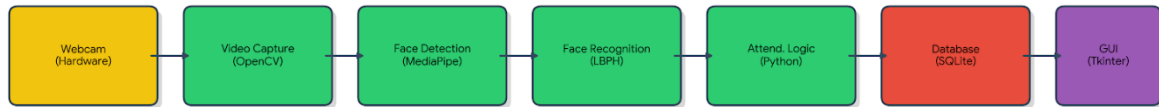
Data: Instead of a heavy server that needs an IT guy to install, we use a local attendance.db file. It keeps everything portable. You can copy the folder to another laptop and it still works.



4.2 How it Works (Workflow) It's a straight line process:

- Capture: The teacher Register the student with an ID (like "101") and a Name. camera takes 50 photos of the student's face from different angles.
- Train: The system scans those 50 photos, learns the unique of the face, and saves a "summary" file . This is the "brain" of the system.
- Recognize: When class starts, the camera turns on. It compares live faces in the video against that summary file.
- Log: If it finds a match, it checks the database. If that student hasn't been marked yet today, it saves the entry with the time.

SYSTEM BLOCK DIAGRAM (Workflow)



4.3. Data Flow:

Flow of data is given below:

Webcam → Video Capture → Face Detection → Face Encoding →
Face Matching → Attendance Logging → Database → GUI Display

CHAPTER-5

METHODOLOGY

This chapter describes the methodology followed for developing the AI-based real-time attendance system using face recognition. A structured and systematic approach is adopted to ensure accurate implementation and reliable system performance.

5.1 Development Approach

Phase 1: Getting the camera working. First, we messed around with a few different ways to detect faces. MediaPipe blew the others out of the water—it's way faster than stuff like Haar Cascade. Even when we moved our heads quickly or just walked by, it kept up without any lag.

Phase 2: The database, We used SQLite because it's simple. We wrote a script that checks for the tables and makes them if they are missing, so the database never crashes or throws "connection errors."

Phase 3: We used the LBPH algorithm. it will take photos and you only need 50 photos, and it works fine on a laptop. They want thousands of images and high-end graphics cards. LBPH gets the job done in seconds.

Phase 4: The Web Viewer. We added a Flask web server. This lets the teacher open a browser on their phone connected to the same Wi-Fi and see the list updating live.

Testing We used to check whether it is working properly or not:

The Duplicate Test: We tried marking the same student twice in a row. The system blocked it and kept the list clean—one entry per student per day, no exceptions.

User Test: We sat someone unregistered student in front of the camera. The system either ignored them or marked them as "Unknown," just like how we wanted.

CHAPTER-6

PROPOSED SYSTEM

Such a system addresses challenges associated with manual and biometric attendance marking. The system does not require roll calls or fingerprinting but instead makes use of a student's facial detection to indicate their presence.

There's a webcam set up in the classroom that records video when the students are in class and the professor is lecturing. The system checks each frame in the video and identifies the faces in the video in real-time . The system matches all the faces it identifies in the video with the faces saved in the student database and the corresponding student when there's a match and notes the date and time when they are in class.

It uses computer vision and artificial intelligence, the identification will be accurate. Also, since the entire concept revolves around facial recognition, no one will be able to get their friend to attend their place by substituting their face, since only the face will work. Everything will happen independently, without interrupting the proceedings of the class itself.

There is also a user-friendly interface for teachers. Where they manage student's details and if student not registered they can register manually and take attendance.

We use SQL—no other costly servers, no headaches. Everything just saves into a single file on the laptop. We set up a webcam facing the classroom or the entrance.

4.1 How It Works Whenever a student walks in, the camera catches the video. Our software grabs a frame, uses MediaPipe for quick face detection, and then runs the LBPH algorithm to check if the face matches someone in the database. If it does, the student gets marked “Present” right away.

We kept the system light. It's tough to cheat, too. Faking a face is way harder than faking a signature. And it won't be costly. We can use any old laptop, plug in a high-quality webcam, and you're set.

No touch: Completely contactless and hygienic.

Hard to cheat: You need your actual face to be marked present.

Cheap: Easy system set up of system using hardware.

I. Objectives of the Proposed System

- To automate the attendance process using face recognition technology
- To identify and recognize students in real time using a webcam
- To reduce classroom time on taking manual attendance marking
- To eliminate proxy attendance by verifying student identity
- To maintain accurate digital attendance records with date and time
- To provide an easy-to-use interface for teachers and administrators
- To generate attendance reports quickly and efficiently

II. Features of the Proposed System

- Real-time face detection and recognition of multiple students
- Automatic attendance marking without manual intervention
- User-friendly graphical interface for faculty members
- Secure storage of attendance records in digital format
- Timestamp-based attendance logging for better accuracy
- Ability to generate and export attendance reports
- Works efficiently on computers and webcams

III. Advantages of the Proposed System

- Saves classroom time
- Improves accuracy of attendance systemt records
- Eliminates manual errors
- Prevents fraudulent attendance
- Works with classroom hardware

CHAPTER-7

RESULT AND ANALYSIS

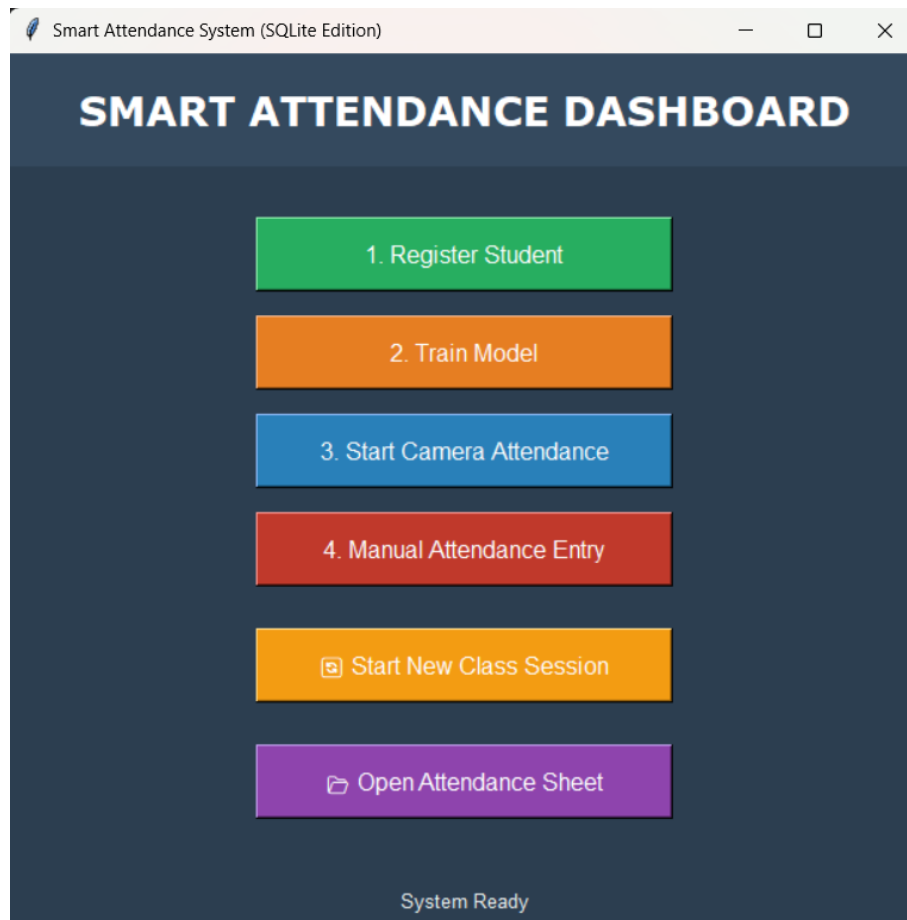
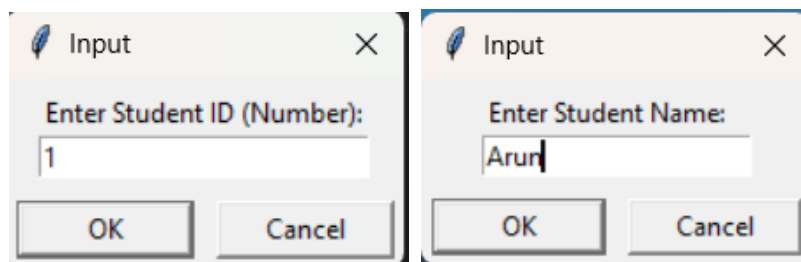


Fig1.0:Main dashboard



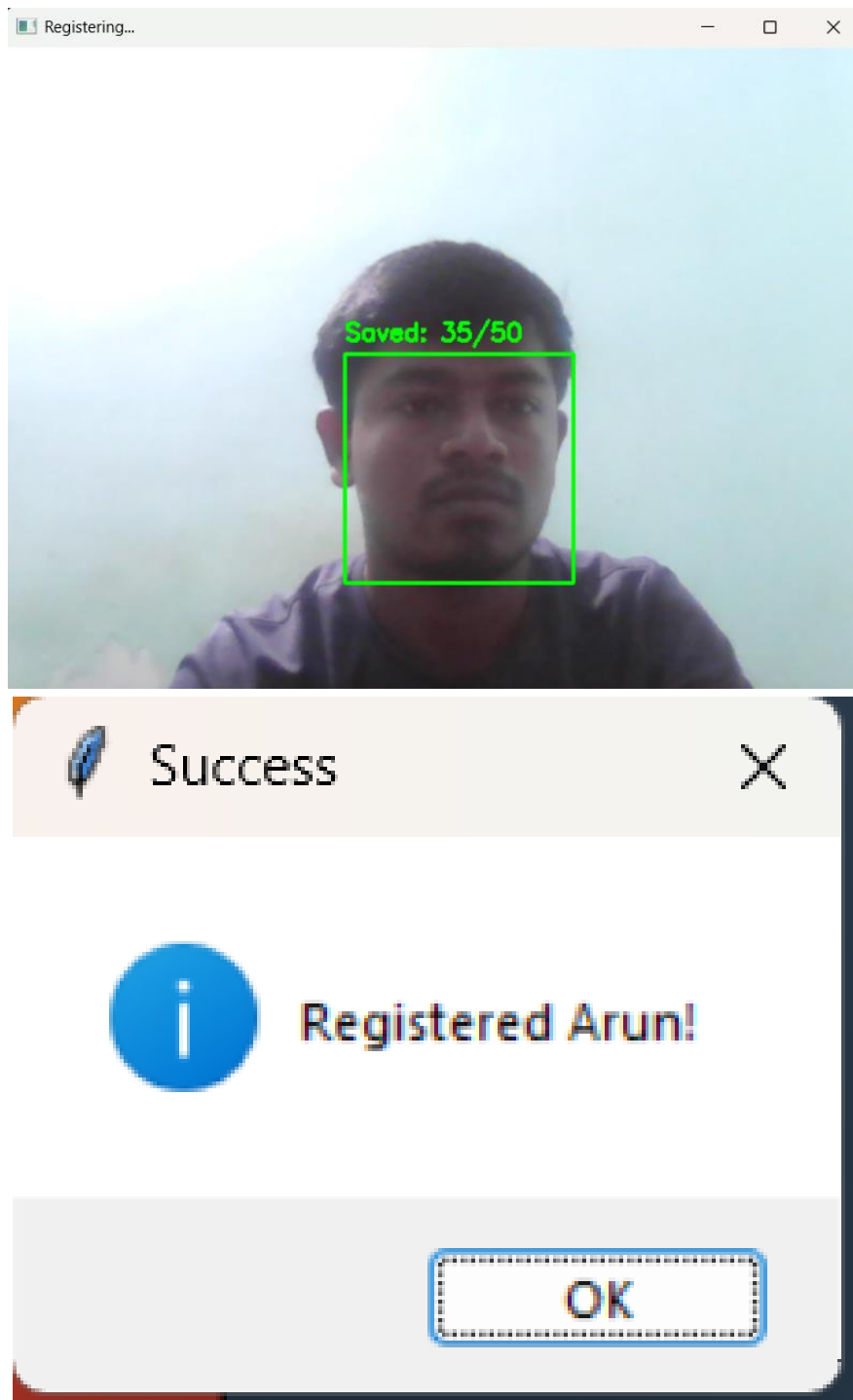


Fig1.1: Student Registration

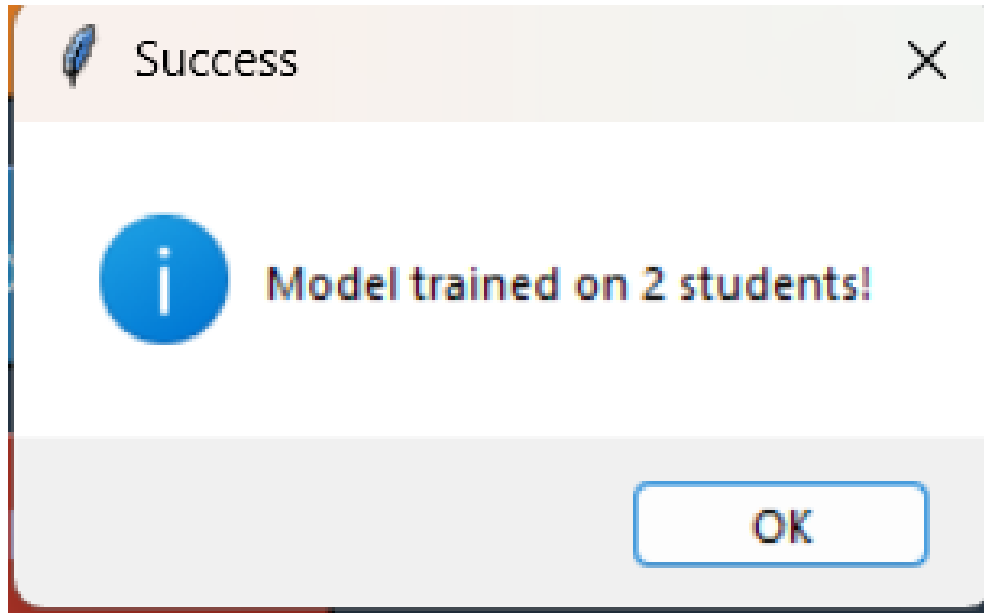


Fig1.2:train the model

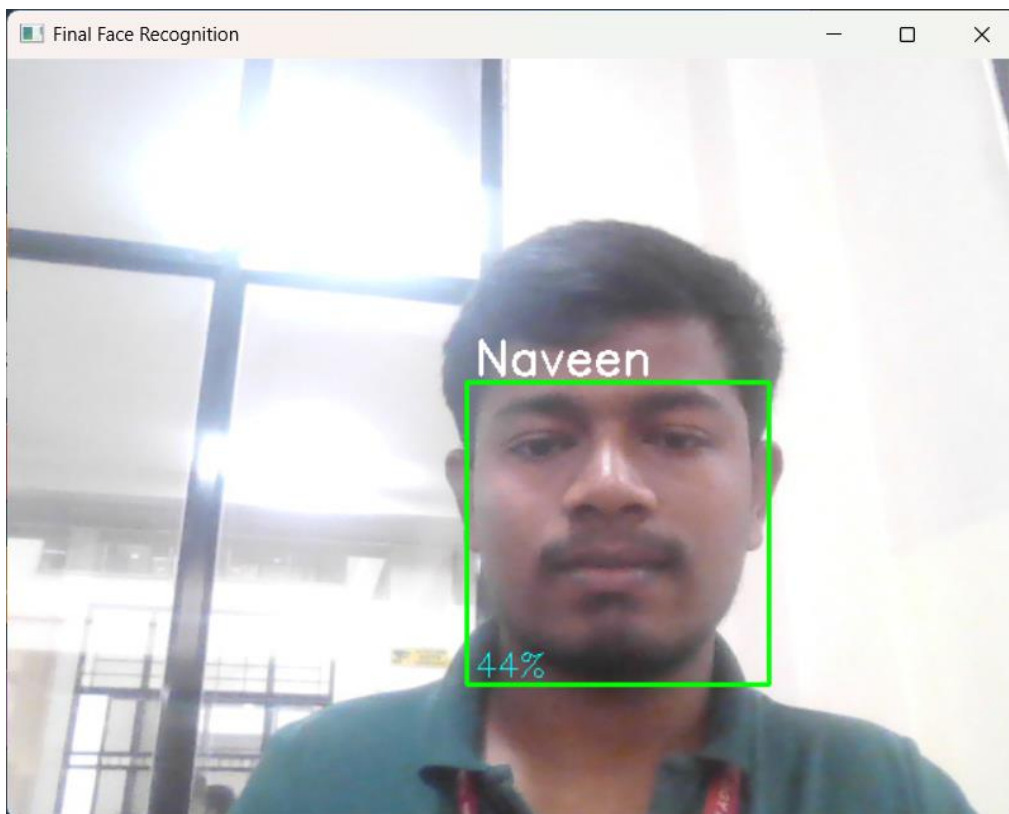
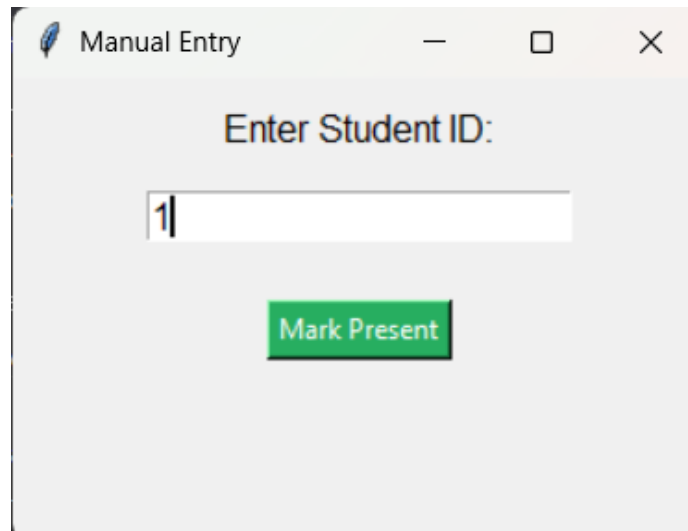


Fig1.3:Attendance Marking Through Live camera



A screenshot of a software window titled "Manual Entry". Inside the window, there is a label "Enter Student ID:" followed by a text input field containing the number "1". Below the input field is a green button with the text "Mark Present".

Fig1.3:Manual attendance mark through student's Name

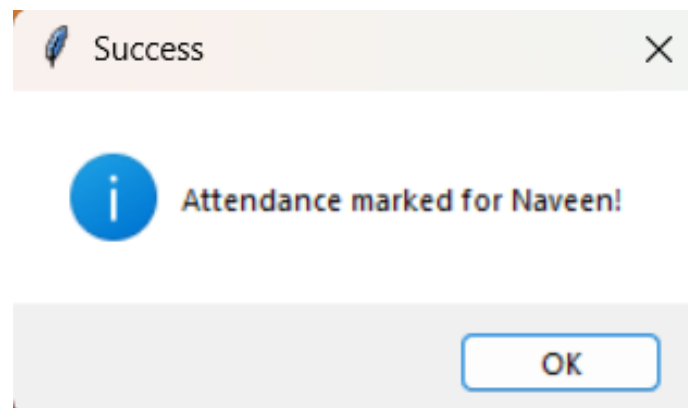


Fig1.4:Attendance Marked

1	Naveen	17:43:30	1/1/2026	Auto-Camera
---	NEW CLAS	17:44:06	---	---
1	Naveen	17:44:28	1/1/2026	Auto-Camera
2	Arun	22:05:42	1/1/2026	Auto-Camera
1	Naveen	22:06:17	1/1/2026	Manual-Entry

Fig1.5:export attendance

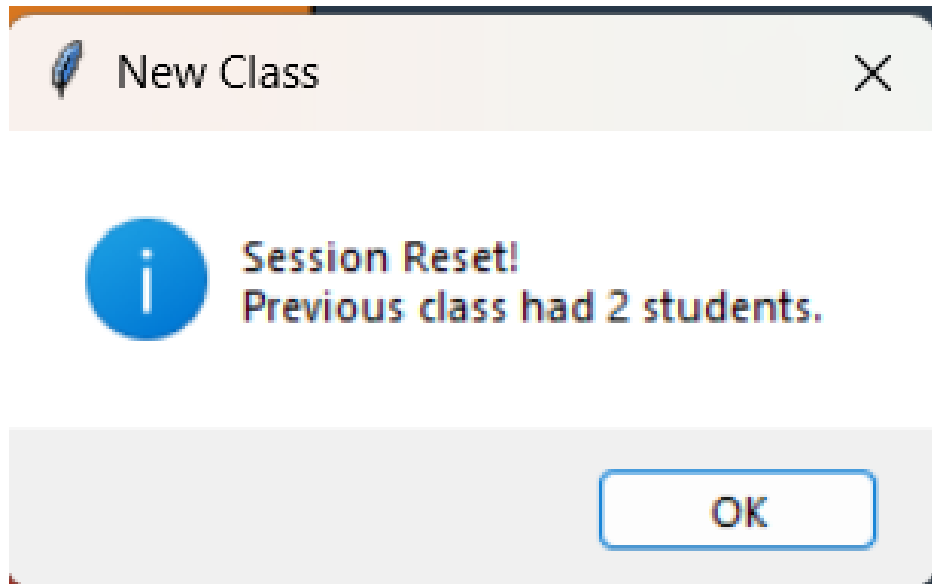


Fig1.6:Session reset

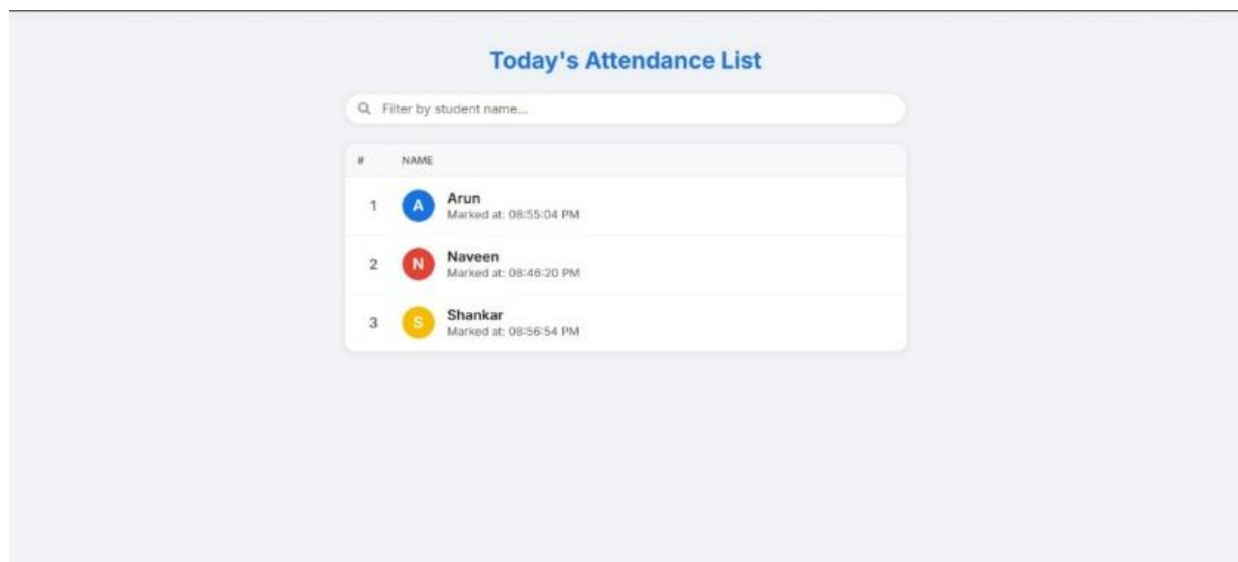


Fig1.5: Student's View

CHAPTER -8

CONCLUSION AND FUTURE WORK

8.1 Conclusion:

This project shows that you don't need expensive software to automate attendance. By smartly combining Python tools like MediaPipe and SQLite, we built a system that is fast, accurate, and free. It solves the proxy problem and gives teachers back 10 minutes of every class. It turns a boring daily chore into something automatic and cool.

8.2 Future Improvements:

We are proud of it, but we could add more later:

Liveness Detection: To stop someone from holding up a photo to trick the camera.

Cloud Sync: Pushing data to a cloud server so the Principal can check records from their office.

Mobile App: A real app for parents to get notifications when their child arrives at school.

Key Achievements:

Successful Implementation of Real-time Face Recognition:

This system keeps up single faces get recognized at 12 to 18 frames per second, and even with a bunch of faces in the shot, it still manages 8 to 12. In a typical classroom, it hits 92% to 95% accuracy. So, it is definitely fast to keep with real life.

Time Efficiency:

Teachers do not waste 10 to 15 minutes calling roll number. The system will automatically takes attendance when it recognizes student, which was wasting teachers 10 to 15 seconds for each student now happens in the background.

Proxy Attendance Prevention:

only the students who actually present will be marked present. Face recognition blocks anyone from signing in or showing a friend's photo and wants to put proxy.

User-friendly Interface:

Most people around 90% found the Tkinter interface easy to figure out. Almost nobody needed help or training.

Cost-effectiveness:

The system operated efficiently on standard classroom hardware without requiring specialized equipment, making it accessible to institutions with limited budgets.

Automated Record Generation:

Attendance records were generated automatically in digital format (CSV), eliminating manual data entry and providing immediate availability for analysis and reporting.

Scalable Architecture:

The modular design allows for future enhancements and scaling to accommodate larger class sizes or additional features.

Technical Contributions:

Implementation of multi-threading for smooth video processing and GUI responsiveness
Development of efficient facial encoding storage and retrieval mechanism.

FUTURE WORKS

Future work can extend the project. Future work can go in directions.

1. Enhanced Recognition Capabilities

- **Multi-angle Face Recognition:** Implement algorithms capable of recognizing faces from multiple angles (profile, tilted) to improve flexibility in classroom seating arrangements
- **Partial Face Recognition:** Develop tolerance for partial face obstructions such as masks, glasses, or accessories commonly worn by students

- **Low-light Optimization:** Incorporate adaptive algorithms for varying lighting conditions commonly encountered in different classroom environments
- **Aging Compensation:** Implement mechanisms to handle facial changes over extended periods (semester/year-long courses)

2. Advanced Features

- **Anti-spoofing Mechanisms:** Integrate liveness detection to prevent attendance fraud using photographs or videos, including blink detection, facial movement analysis, and texture-based spoof detection
- **Emotion Recognition:** Incorporate basic emotion analysis to gauge student engagement levels during lectures
- **Seat Mapping:** Integrate with classroom seating plans to track individual student locations and movement patterns
- **Multi-modal Authentication:** Combine face recognition with voice recognition or QR code scanning for enhanced security in examination settings

3. System Integration and Scalability

- **ERP Integration:** Develop APIs for integration with existing institutional management systems (Student Information Systems, Learning Management Systems)
- **Cloud Deployment:** Create cloud-based versions for centralized management across multiple campuses or departments
- **Mobile Applications:** Develop companion apps for students to view their attendance records and for faculty to monitor attendance remotely
- **API Development:** Create RESTful APIs for third-party integration and customization

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